Practical 11: Forecasting Nottem Dataset

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1. Question

Consider the "nottem" dataset available in R-Library. Forecast the temperature values for the next two years.

2. Objective

To perform time series modeling and forecasting on the Nottingham temperature dataset using ARIMA methodology. The objective is to examine the stationarity of the dataset, identify the appropriate ARIMA model, estimate the model parameters, check the adequacy of the model using diagnostic tests, and forecast future temperature values.

3. R Code

```r
# Remove previously stored variables
rm(list = ls(all = TRUE))

# (a) Load dataset from R library
data("nottem")

# (b) Convert dataset into time series object
temperature_ts <- as.ts(nottem)

# Print dataset
temperature_ts

# (c) Plot the data to identify dominant component
plot(
  temperature_ts,
  main = "Nottingham Monthly Temperature Data",
  xlab = "Time",
  ylab = "Temperature",
  col = "blue",
  lwd = 2
)
```

```r
# (d) Load required packages for time series analysis
library(tseries)
library(forecast)

# (e) Check stationarity using ACF and PACF
acf(
  temperature_ts,
  main = "ACF of Nottingham Temperature",
  lag.max = 20,
  ylim = c(-1,1)
)
```

```r
pacf(
  temperature_ts,
  main = "PACF of Nottingham Temperature",
  lag.max = 20,
  ylim = c(-1,1)
)
```

```r
# (f) Stationarity test using KPSS Test
kpss.test(temperature_ts)
```

```
## Warning in kpss.test(temperature_ts): p-value greater than printed p-value
```

```r
# (g) If non-stationary, make it stationary using differencing
temperature_d1 <- diff(temperature_ts)

kpss.test(temperature_d1)
```

```
## Warning in kpss.test(temperature_d1): p-value greater than printed p-value
```

```r
plot(
  temperature_d1,
  main = "First Differenced Series",
  xlab = "Time",
  col = "blue",
  lwd = 2
)
```

```r
acf(
  temperature_d1,
  main = "ACF of Differenced Series",
  lag.max = 20,
  ylim = c(-1,1)
)
```

```r
pacf(
  temperature_d1,
  main = "PACF of Differenced Series",
  lag.max = 20,
  ylim = c(-1,1)
)
```

```r
# (h) Model Identification and Parameter Estimation using ARIMA
temperature_fit <- auto.arima(temperature_ts)

temperature_fit

# (i) Diagnostic Checking (Goodness of Fit)
res <- temperature_fit$residuals

# Plot residuals
plot(
  res,
  main = "Residuals of Nottingham Temperature Model"
)
```

```r
# Test 1: Normality Test
hist(
  res,
  main = "Histogram of Residuals"
)
```

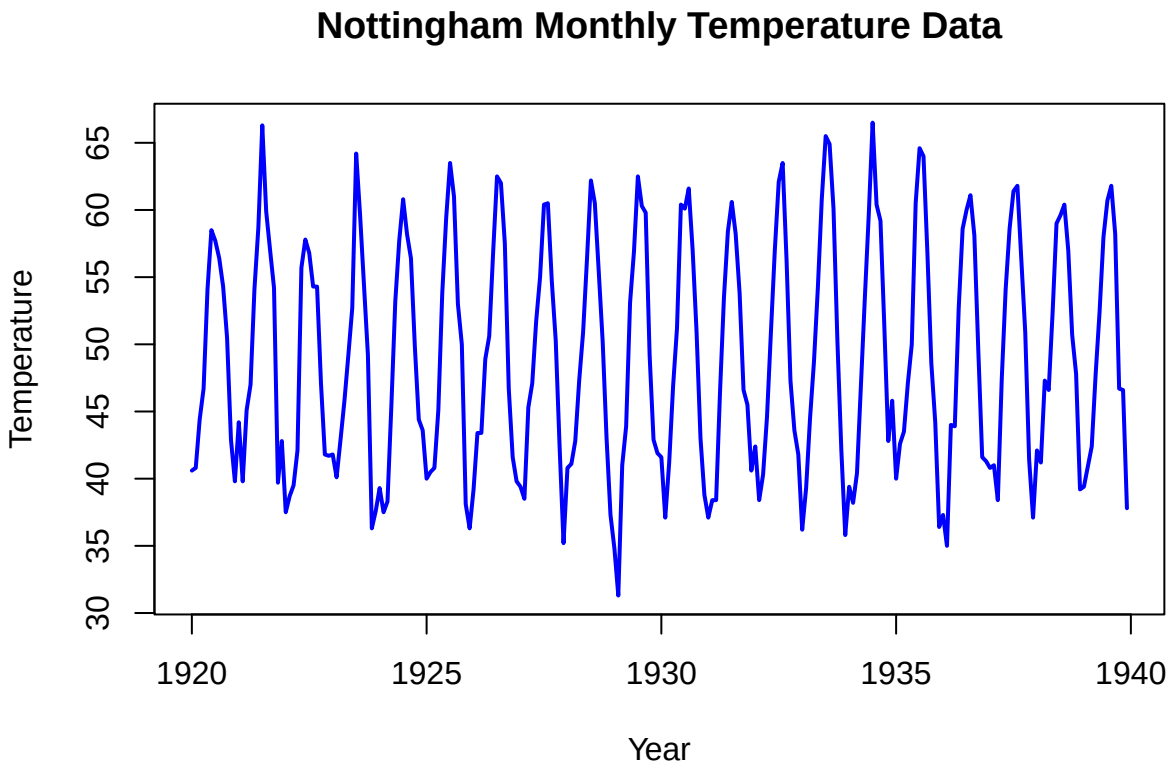
```r
shapiro.test(res)

# Test 2: Serial Correlation Test
Box.test(
  res,
  lag = 20,
  type = "Box-Pierce"
)

# Test 3: Heteroscedasticity Check
acf(
  res^2,
  main = "ACF of Squared Residuals"
)
```

```r
pacf(
  res^2,
  main = "PACF of Squared Residuals"
)
```

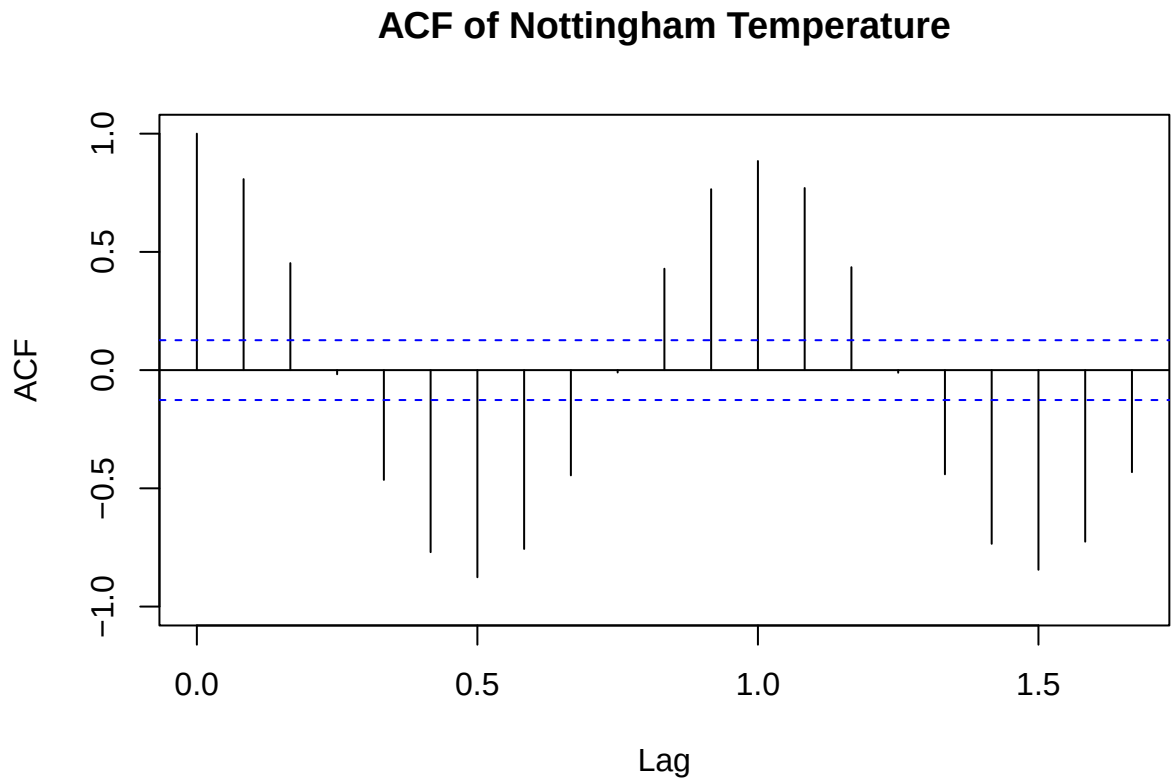
```r
# (j) Forecast next two years (24 months)
temperature_forecast <- forecast(
  temperature_fit,
  h = 24
)

temperature_forecast

plot(
  temperature_forecast,
  main = "Forecast of Nottingham Temperature for Next 2 Years"
)
```

4. Output

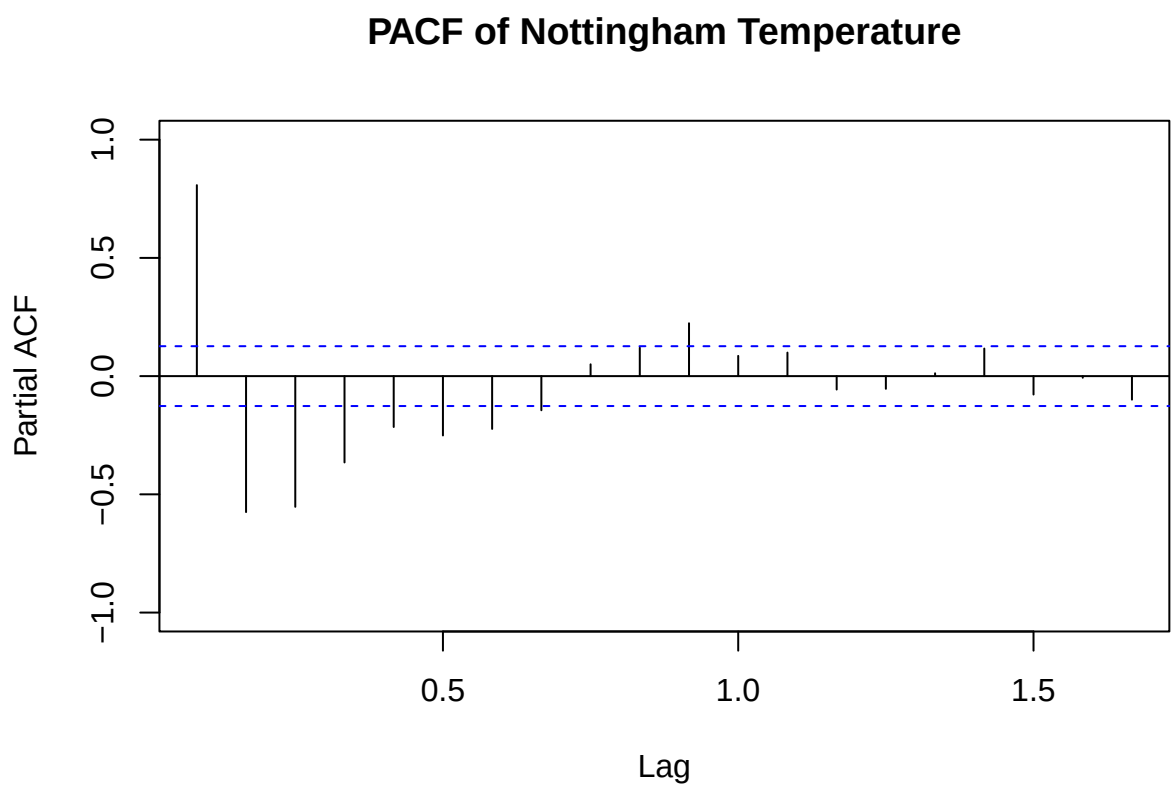
Time Series Plot



ACF Plot


ACF of Nottingham Temperature
ACF
1.0
0.5
0.0
−0.5
−1.0
0.0
0.5
1.0
1.5
Lag

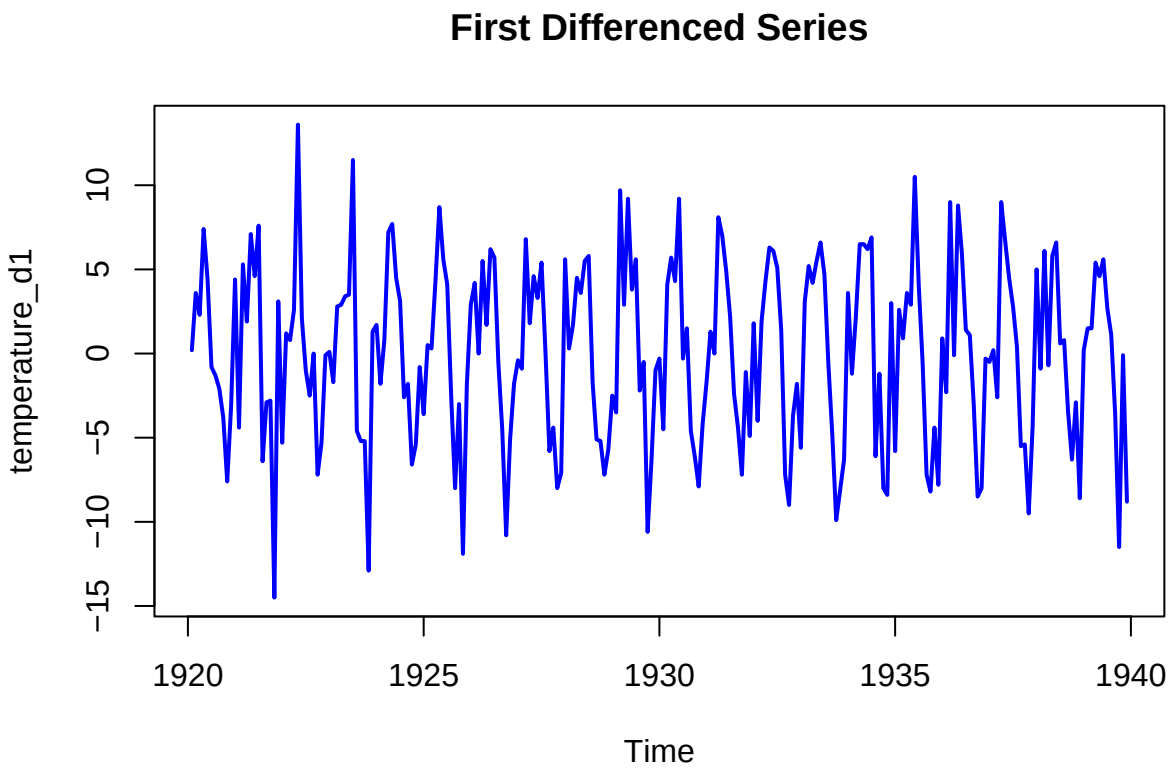
PACF Plot



KPSS OUTPUT

```
## Warning in kpss.test(temperature_ts): p-value greater than printed p-value

##
##  KPSS Test for Level Stationarity
##
## data:  temperature_ts
## KPSS Level = 0.032053, Truncation lag parameter = 4, p-value = 0.1
```

Differenced Series



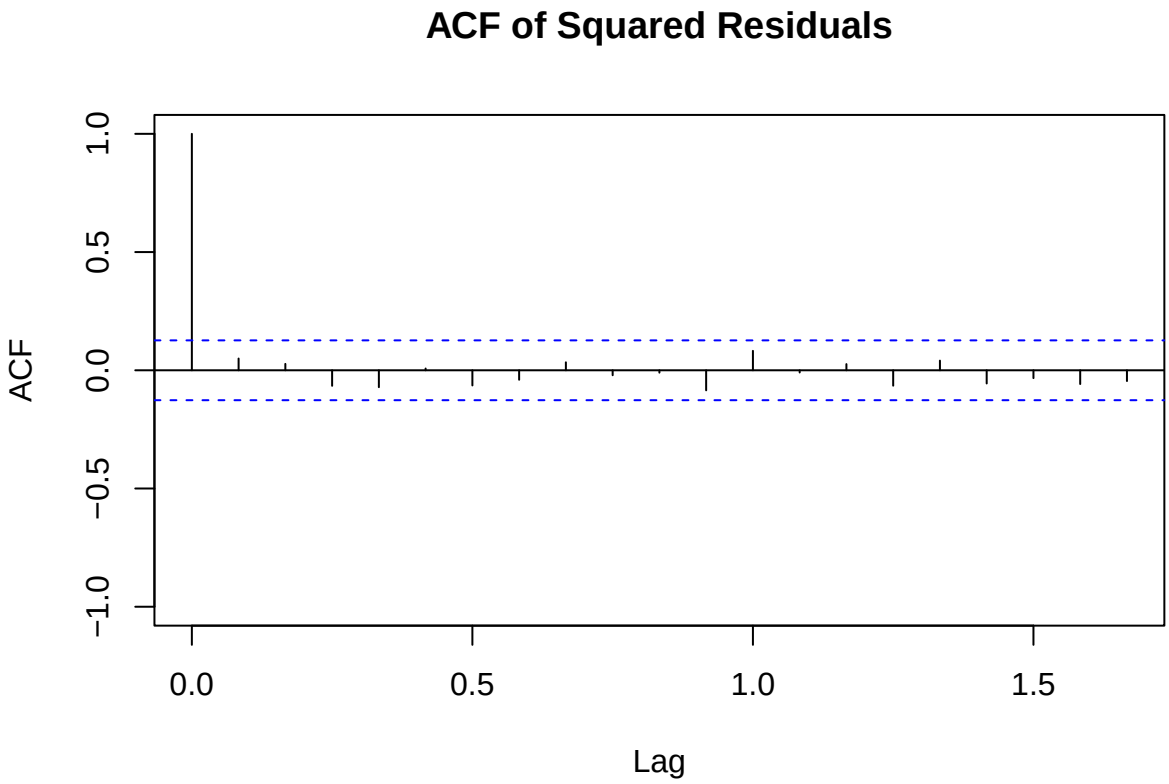
###Shapiro-Wilk Normality Test

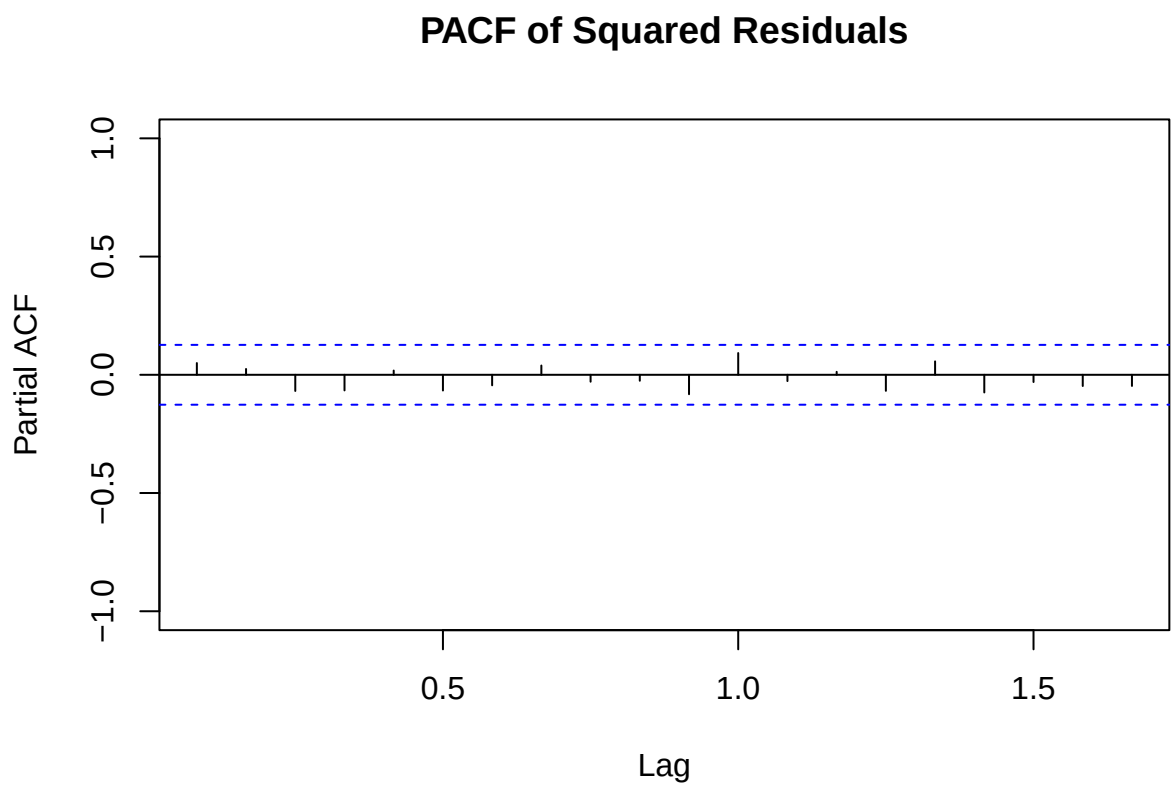
```
##
##  Shapiro-Wilk normality test
##
## data:  res
## W = 0.99232, p-value = 0.2461
```

###Box-Pierce Serial Correlation Test

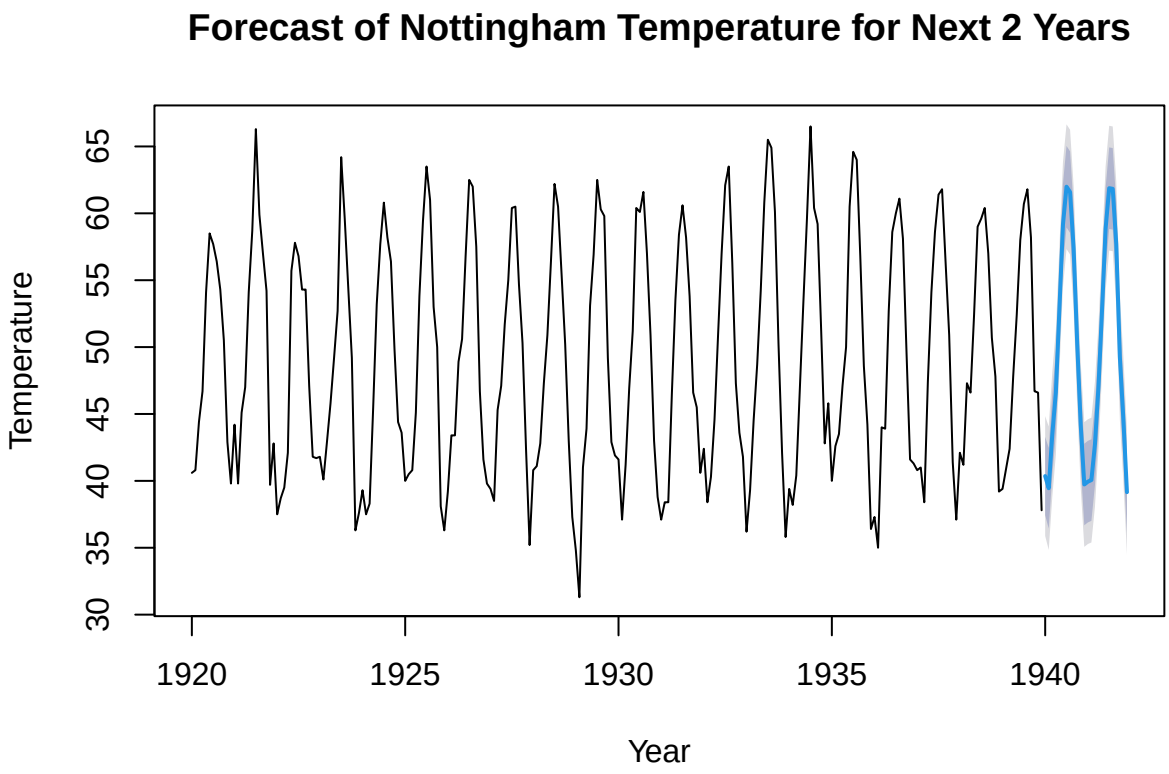
```
##
##  Box-Pierce test
##
## data:  res
## X-squared = 14.972, df = 20, p-value = 0.778
```

heteroscedasticity test





##Forecast Plot



5. Conclusion

The Nottingham temperature dataset was successfully converted into a time series object and analyzed using ARIMA modeling techniques.

The time series plot indicated the presence of strong seasonal variation in the temperature data. ACF and PACF plots suggested the presence of correlation in the series, and the KPSS test confirmed that the original series was non-stationary.

After applying first differencing, the series became stationary. Based on the structure of the series and AIC criterion, an ARIMA model was selected using the auto.arima() function, and the model parameters were estimated.

Diagnostic checking confirmed that:

• Residuals are approximately normally distributed (Shapiro-Wilk test) • Residuals show no serial correlation (Box-Pierce test) • Residual variance appears constant based on ACF/PACF of squared residuals

Hence, the selected ARIMA model provides an adequate fit for the Nottingham temperature dataset. Using the fitted model, temperature values were forecasted for the next 24 months, indicating the continuation of the seasonal pattern in future observations.